

Food Ordering Behaviour and Consumer Trends

Final Project Report

Project Detail	Information
Project Title	Food Ordering Behaviour and Consumer Trends
Project Type	Individual Project
Analysis Platform	Tableau Desktop
Project Duration	1 August 2026 – 28 August 2026
Dataset	food_ordering_behavior_dataset.csv
Dataset Size	50,000 records and 19 fields
Tableau Workbook	Food.twb
GitHub Repository	https://github.com/25eg105h26-code/Food-Ordering-Behaviour-and-Consumer-Trends

Data Analysis with Tableau

Project Initialization, Planning, Visualization & Insights

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1. Introduction

1.1 Project Overview

The food delivery industry is becoming increasingly competitive. Businesses must understand changing customer preferences, improve operational efficiency, provide satisfactory service, and maximize revenue. Although food-ordering platforms generate large volumes of data, raw records alone do not clearly reveal the factors influencing order demand, customer spending, cuisine preferences, or service performance.

This project, titled **Food Ordering Behaviour and Consumer Trends**, uses Tableau to transform a food-ordering dataset into meaningful visual insights. The supplied dataset contains **50,000 order records** and **19 fields**, including customer characteristics, city, cuisine, meal type, restaurant type, order value, delivery fee, customer rating, repeat-order behaviour, mood, hunger level, company, rainy-weather conditions, and time taken to order.

The project follows a structured data-analysis process:

1. Define the business problems and stakeholder requirements.
2. Identify and validate the source dataset.
3. Assess data quality and prepare the data for analysis.
4. Frame relevant business questions.
5. Develop visualizations in Tableau.
6. Combine the visualizations into an interactive dashboard.
7. Present the findings through a Tableau story.
8. Derive business, operational, and customer-focused conclusions.

The analysis helps identify important patterns such as the highest-performing cities, customer age-group behaviour, cuisine preferences, meal-type performance, order-processing time distribution, customer ratings, and spending patterns. Interactive filters allow decision-makers to examine these findings by city, cuisine, and meal type.

1.2 Objectives

The primary objective is to analyze food-ordering behaviour and consumer trends using Tableau and convert the available raw data into actionable business information. The specific objectives are:

- To verify and prepare the dataset containing 50,000 records and 19 fields.
- To summarize business performance through total orders, order value, delivery fees, and average customer ratings.
- To identify cities that generate the highest order volumes.
- To compare customer demand across cuisines, meal types, restaurant types, and age groups.
- To examine customer spending, ratings, and repeat-order behaviour.
- To evaluate the distribution of time_taken_to_order.
- To identify peak meal periods and workload patterns affecting operations.
- To develop an interactive Tableau dashboard with relevant filters.
- To construct a Tableau story that communicates important consumer and business trends.
- To support strategic, operational, and customer-experience decisions with visual evidence.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

The project considers three stakeholder scenarios: a business analyst, an operations manager, and a customer.

PS	Stakeholder	Trying To	Barrier	Reason	Feeling
PS-1	Rahul Sharma, Business Analyst	Understand order trends, cuisine, city demand, and spending.	Raw data is difficult to compare quickly.	Many fields hide the strongest patterns.	Unsure about key priorities.
PS-2	Priya Mehta, Operations Manager	Improve service efficiency using order-time and city patterns.	Busy periods and workload gaps are hard to spot.	Raw records do not show operational patterns clearly.	Concerned about service gaps and satisfaction.
PS-3	Neha Reddy, Customer	Get quick delivery, reasonable prices, and varied cuisine choices.	Delivery delays or limited choices can affect the experience.	Service quality and convenience influence repeat orders.	Satisfied when orders are timely, affordable, and convenient.

Rahul Sharma – Business Analyst

Rahul analyzes order trends, cuisine preferences, city-wise demand, and customer spending. He uses Tableau dashboards to identify high-performing segments. The principal problem is that raw records do not make comparisons immediately visible.

Priya Mehta – Operations Manager

Priya focuses on operational efficiency. She examines time_taken_to_order, city order volumes, and meal-period workloads. An interactive operational view helps identify peak demand and improve resource allocation.

Neha Reddy – Customer

Neha represents a regular customer who values timely service, reasonable pricing, and varied cuisine. Satisfaction depends on speed, cost, choice, and quality—factors that influence repeat orders and loyalty.

2.2 Project Proposal (Proposed Solution)

The proposed solution is an interactive Tableau analysis that converts the food-ordering dataset into accessible visual insights for business, operational, and customer-focused decisions.

Component	Description
Objective	Analyze 50,000 records to identify demand, spending, cuisine, meal, city, satisfaction, and operational patterns.
Scope	Descriptive analysis of city, cuisine, meal type, order time, restaurant type, order value, delivery fee, ratings, repeat orders, and time_taken_to_order.
Approach	Import and validate CSV, create calculated summaries, develop charts, apply filters, build dashboard and Tableau story.
Solution File	Food.twb
Expected Result	Faster identification of high-demand markets, preferences, spending patterns, workload distribution, and customer-experience indicators.

Key Features

- KPI cards for total orders, order value, delivery fees, and average rating.
- City-wise order-volume ranking and cuisine-preference distribution.
- City–cuisine heat map; age-group and restaurant-type comparison.
- Company-type and meal-type analysis; order-processing-time distribution.
- Interactive city, cuisine, and meal-type filters; Tableau story points.

Resource Requirements

Resource	Specification
Computing device	Laptop/desktop, 4-core CPU or equivalent
Memory / Storage	8 GB RAM or higher; 256 GB SSD or higher
Software	Tableau Desktop; Food.twb workbook
Data source	food_ordering_behavior_dataset.csv – 50,000 rows, 19 columns

2.3 Initial Project Planning

Sprint	Epic	USN	Task	Pts	Schedule
Sprint 1	Data Preparation	USN-1	Load CSV into Tableau; verify 50,000 records and 19 fields.	3	1–8 Aug 2026
Sprint 1	Exploratory Analysis	USN-2	Analyze city, cuisine, meal type, and order-time trends.	3	1–8 Aug 2026

Sprint	Epic	USN	Task	Pts	Schedule
Sprint 2	Customer & Revenue	USN-3	Compare order value, delivery fee, ratings, and repeat orders.	3	9–18 Aug 2026
Sprint 2	Operations Analysis	USN-4	Analyze time_taken_to_order and city/time workloads.	3	9–18 Aug 2026
Sprint 3	Dashboard & Insights	USN-5	Develop interactive dashboards and communicate key findings.	2	19–28 Aug 2026

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The project uses one supplied CSV file as its primary analytical source. No external data collection was required.

Section	Description
Project Overview	Analyze food-ordering behaviour and performance using Tableau.
Collection Method	Use the supplied CSV file and connect it directly to Tableau.
Validation Method	Verify row/field counts, missing values, duplicates, numeric ranges, categorical consistency.
Analysis Areas	City, cuisine, meal type, restaurant type, spending, delivery fee, rating, behaviour, order-processing time.

Raw Data Source

Source	Description	Format	Size	Access
Dataset 1	50,000 food orders with customer, city, cuisine, meal, spending, fee, rating and behaviour fields.	CSV	~5.07 MB	Provided for project use

3.2 Data Quality Report

Check	Finding	Severity	Resolution
Missing values	All 19 fields contain zero missing values.	Low	Run null checks before every refresh.
Duplicates	No duplicate rows among 50,000 records.	Low	Retain all validated records.
Numeric ranges	Values within valid analytical ranges.	Low	Apply range checks during refreshes.
Field naming	time_taken_to_order may be confused with delivery time.	Moderate	Document meaning; use clear display label.
ID aggregation	Order ID / User ID may be summed incorrectly.	Moderate	Set as identifiers/dimensions; avoid SUM.

The dataset was suitable for descriptive analysis without deletion, imputation, or major restructuring.

3.3 Data Exploration and Preprocessing

The dataset contains 50,000 records and 19 fields covering customer characteristics, order identifiers, location, cuisine, meal type, restaurant type, spending, delivery fees, ratings, repeat-order indicators, mood, hunger level, company, rainy-weather conditions, and order-processing time.

Cleaning: confirmed 50,000 records / 19 fields; zero missing values; zero duplicates; valid numeric ranges; categorical consistency reviewed; Order ID and User ID treated as identifiers.

Transformations: Age Group calculated field; Order ID aggregated as count; Order Value and Delivery Fee as total/average; average rating; sorting and grouping by city, cuisine, meal type, company type, restaurant type; time_taken_to_order distribution analysis.

Field Category	Tableau Treatment
Identifiers	Order ID and User ID as dimensions/identifiers.
Numeric measures	Age, Order Value, Delivery Fee, Rating, Time Taken to Order as measures.
Categorical dimensions	City, Cuisine, Meal Type, Restaurant Type, Company Type, Mood as dimensions.
Calculated	Age Group created from numeric age field.

A single flat-file model is used; no joins or table relationships are required.

4. Data Visualization

4.1 Framing Business Questions

The visual analysis addresses the following stakeholder-focused questions:

1. Which KPIs summarize current food-ordering performance?
2. Which age groups place the most orders across different restaurant types?
3. Which cuisines are most preferred by customers?
4. How do cuisine preferences vary across cities?
5. How do order volumes vary by company type and meal type?
6. Which cities generate the highest order volumes?
7. How is time_taken_to_order distributed?
8. Which meal types create the greatest workload or order-value contribution?
9. How does customer satisfaction vary across meal types?

4.2 Developing Visualizations

The following visualizations were developed in Tableau and are shown with screenshots below.

KPI Summary Cards

KPI cards summarize Total Orders (50,000), Order Value (27.39M), Delivery Fee (2.98M), and Average Rating (2.99).

Orders by Age Group & Restaurant Type

Stacked bar chart comparing order volume by Age Group and Restaurant Type. Millennials and Gen Z dominate order volume.

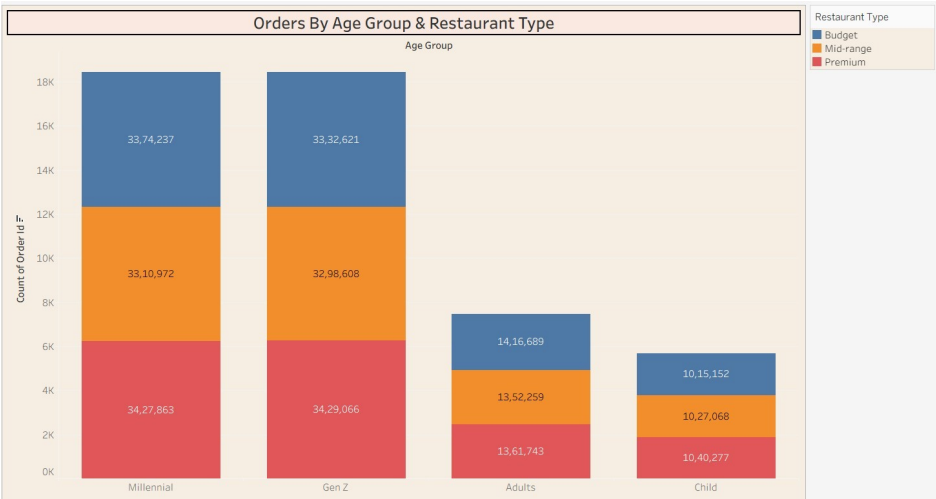


Figure 1: Orders by Age Group & Restaurant Type (Budget / Mid-range / Premium)

Cuisine Preference Distribution

Pie chart showing balanced cuisine demand: each cuisine contributes approximately 16.5%–16.9% of orders.

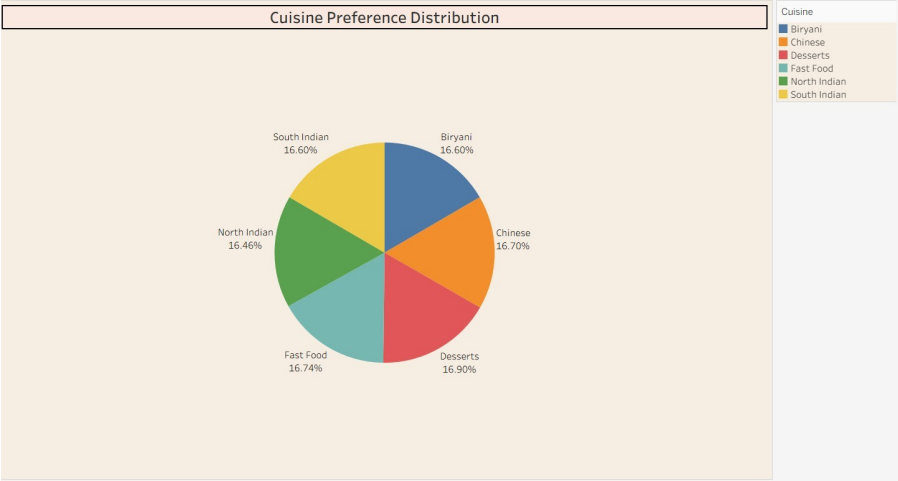


Figure 2: Cuisine Preference Distribution

City vs Cuisine Heat Map

Heat map of order counts by City and Cuisine, supporting city-level menu and promotion planning.

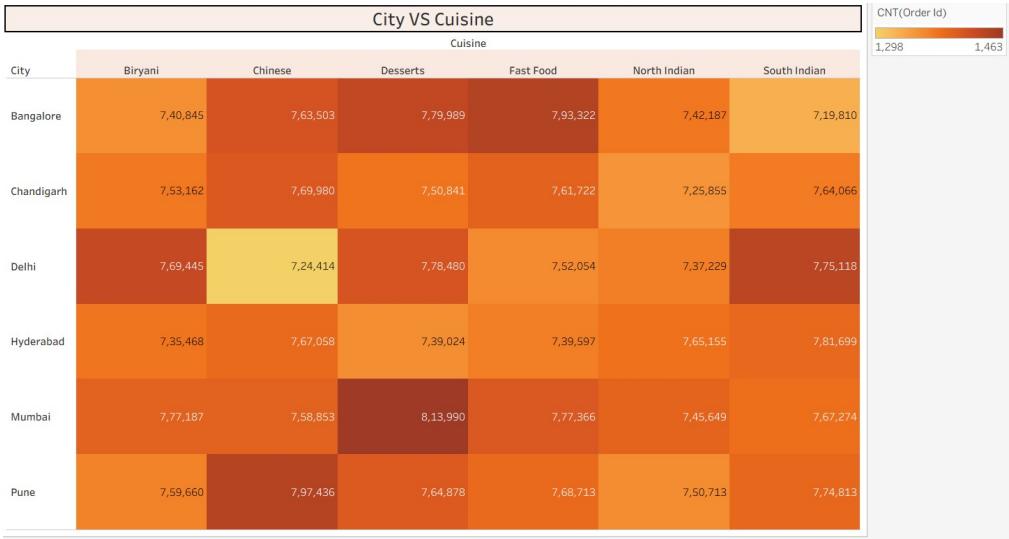


Figure 3: City vs Cuisine Heat Map

Company Type & Meal Type Orders

Bar chart comparing order volumes across Alone, Family, Friends, and Partner segments for selected meal types.

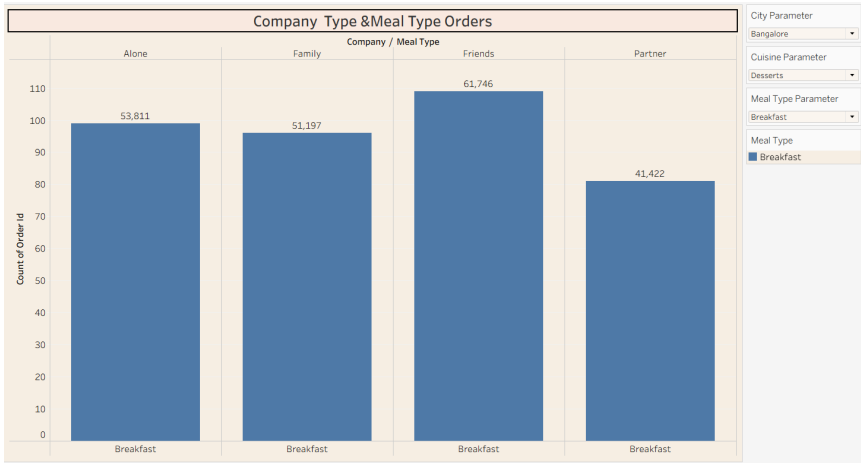


Figure 4: Company Type & Meal Type Orders

Top 5 Cities by Order Volume

Bar chart ranking cities by total order count. Mumbai leads with 8,443 orders, followed by Pune, Bangalore, Delhi, and Chandigarh.

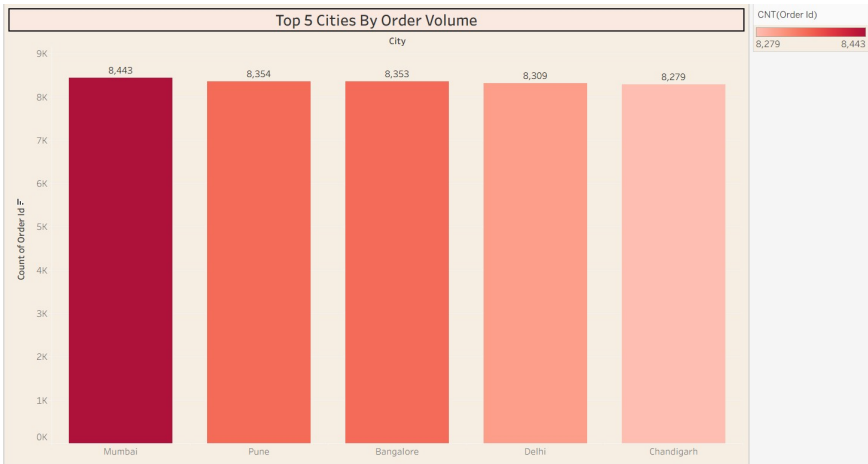


Figure 5: Top 5 Cities by Order Volume — Mumbai leads with 8,443 orders

Additional Operational & Revenue Views

Further charts cover time_taken_to_order distribution, meal-type revenue contribution, and meal type vs customer rating.

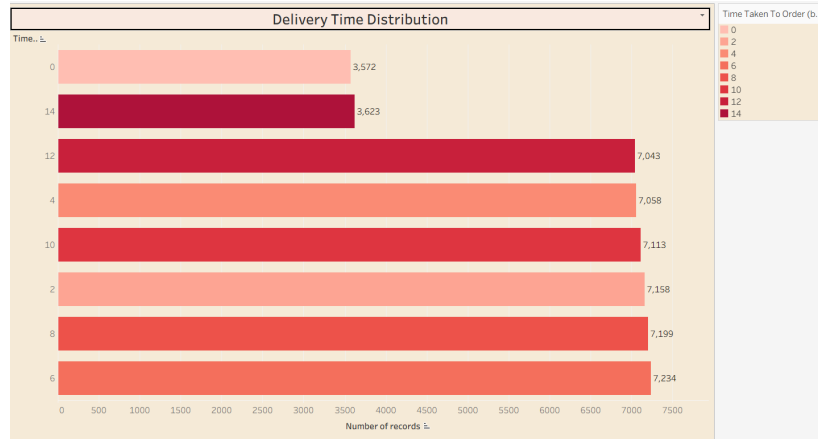


Figure 6: Supporting visualization from the Business Questions report

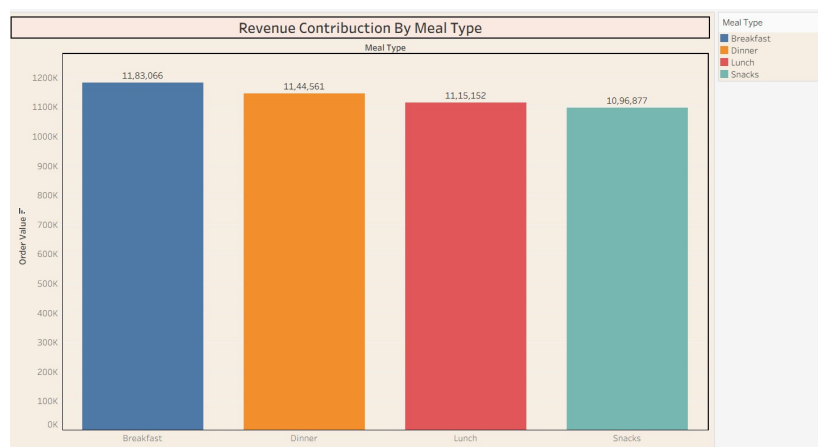


Figure 7: Supporting visualization from the Business Questions report

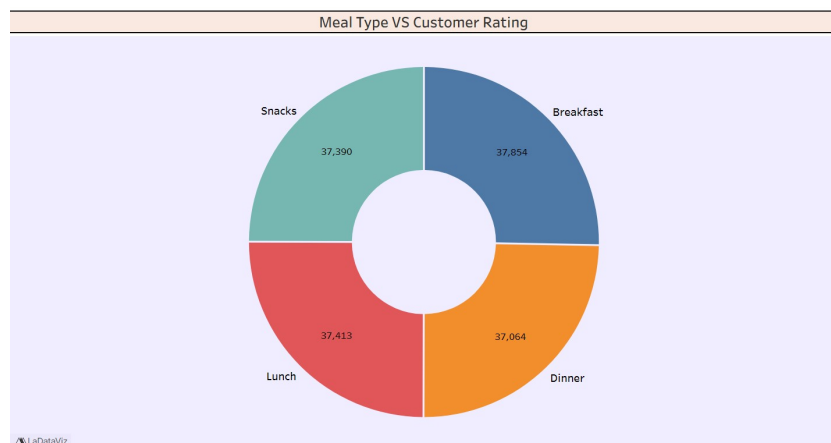


Figure 8: Supporting visualization from the Business Questions report

Visualization Principles Applied

- Bar charts for ranking and category comparisons.
- Donut/pie charts for proportional cuisine distribution.
- Heat map for two-dimensional city–cuisine comparison.
- KPI cards for headline performance measures.
- Consistent colors, data labels, and interactive filters.

5. Dashboard

5.1 Dashboard Design File

The Food Ordering Behavior Dashboard combines the principal Tableau visualizations into one interactive analytical interface. KPI cards sit at the top; detailed charts follow in clearly separated sections.

Full Dashboard Screenshot

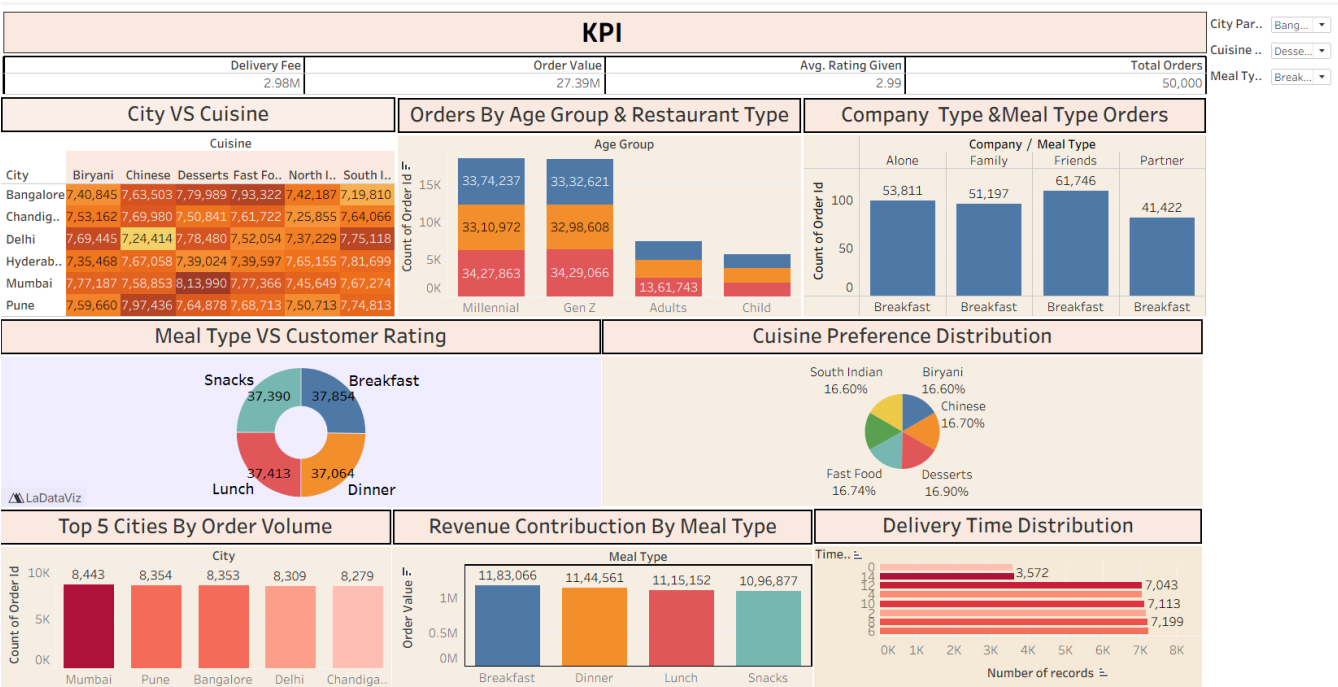


Figure 9: Food Ordering Behavior Dashboard — KPIs, City–Cuisine heatmap, Age Group, Company Type, Cuisine Distribution, Top Cities, Revenue by Meal Type, Delivery Time

Dashboard Design Features

- **Clear layout:** KPI cards first, then detailed charts.
- **Appropriate charts:** Bars, heat map, donut, distribution.
- **Consistent theme:** Warm background and repeated chart colors.
- **Interactive filters:** City, Cuisine, and Meal Type.
- **Drill-down:** Age groups, restaurant/company types, meal types.
- **Emphasized information:** KPI cards, data labels, percentage labels.

Dashboard KPI Results

KPI	Result
Total Orders	50,000
Total Order Value	27.39M
Total Delivery Fees	2.98M
Average Rating	2.99

Major Outcomes

- Mumbai has the highest order volume (8,443 orders) among the top five cities.
- Breakfast has the highest displayed order value (11,83,066), followed by Dinner (11,44,561).
- Cuisine preferences are closely distributed (~16.6%–16.7% each).
- Millennials and Gen Z are the largest age groups by order volume.

6. Report

6.1 Story Design File

The Tableau story **Food Ordering Behaviour and Consumer Trends** presents the analysis as a guided sequence of visual findings covering cuisine, age groups, company/restaurant types, meal types, ratings, city order volume, revenue, and processing time.

Story Screenshot

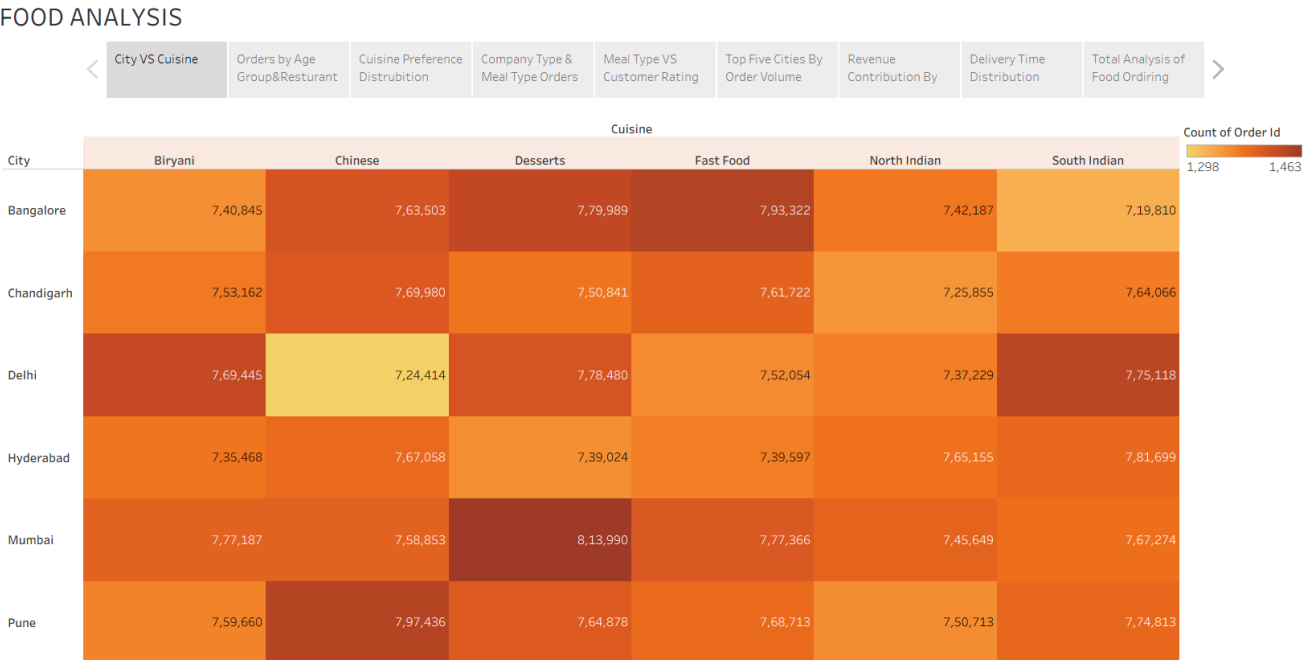


Figure 10: Tableau Story — City vs Cuisine heatmap and navigation across analysis points

Suggested Story Sequence

1. **Business Overview:** Total orders, order value, delivery fees, average rating.
2. **Customer Segmentation:** Age-group demand across restaurant types.
3. **Cuisine Preferences:** Overall distribution and city-level variations.
4. **Customer/Company Behaviour:** Order volume by company and meal type.
5. **City Performance:** Ranking of cities by total order count.
6. **Operational Timing:** Distribution of time_taken_to_order.
7. **Meal-Type Performance:** Workload and order-value contribution.
8. **Customer Satisfaction:** Average ratings across meal types.
9. **Final Findings:** Summary of significant trends and recommended actions.

Key Story Observations

- **Total Orders & Value:** 50,000 orders; order value 27.39M; delivery fees 2.98M.
- **Top City:** Mumbai – 8,443 orders.
- **Cuisine:** Balanced shares (~16.6%–16.7%) across Biryani, Chinese, Desserts, Fast Food, South Indian.
- **Meal Revenue:** Breakfast 11,83,066; Dinner 11,44,561.

7. Performance Testing

7.1 Utilization of Data Filters

Filter	Purpose
City	Focuses the dashboard on a selected market or location.
Cuisine	Isolates customer demand for a selected cuisine.
Meal Type	Compares breakfast, lunch, snacks, and dinner performance.

Filters were validated: selections update charts; KPIs reflect segments; multiple filters combine correctly; reset returns full dataset.

7.2 Number of Calculated Fields

Calculated Field	Purpose
Age Group	Converts numeric age into customer segments.
Order Count	Counts Order ID without summing the identifier.
Cuisine / City / Meal Filter Calculations	Support interactive dashboard filtering.

Five principal calculated/analytical fields are documented.

7.3 Number of Visualizations

No.	Visualization
1	KPI summary cards
2	Age Group by Restaurant Type
3	Cuisine Preference
4	City vs Cuisine Heat Map
5	Company Type by Meal Type
6	Top Cities by Order Volume
7	Time Taken to Order Distribution
8	Meal-Type Workload / Order-Value Comparison
9	Meal Type vs Customer Rating

Performance validation confirmed responsiveness, readability, visual consistency, correct filter behaviour, and proper aggregation. The flat dataset avoids join overhead; identifiers are dimensions to prevent misleading sums.

8. Conclusion and Observations

The project successfully transforms a large food-ordering dataset into an interactive Tableau analysis supporting business, operational, and customer-focused decisions. The dashboard simplifies 50,000 records and 19 fields into interpretable KPIs, comparisons, distributions, and segment-level views.

8.1 Overall Performance

The dashboard reports **50,000 total orders**, order value of **27.39M**, and delivery fees of **2.98M**. Average customer rating is **2.99**, indicating moderate satisfaction and scope to improve customer experience.

8.2 City-Level Demand

Mumbai records the highest order volume (8,443). It is a priority market for resource allocation, partnerships, and promotions. The city filter enables comparison across all markets.

8.3 Cuisine Preferences

Displayed cuisines have similar shares (~16.6%–16.7%). Customers value variety; the city–cuisine heat map supports local menu planning.

8.4 Customer Segments

Millennials and Gen Z are the largest groups by order volume—important for digital campaigns, loyalty strategies, and convenience-focused improvements.

8.5 Meal-Type Performance

Breakfast leads order value (11,83,066), followed by **Dinner** (11,44,561). High-value periods may justify extra capacity; lower periods may benefit from targeted offers.

8.6 Operational Findings

The `time_taken_to_order` distribution helps identify common processing ranges and outliers. Combined with city and meal filters, it reveals whether timing issues concentrate in specific markets or periods.

8.7 Customer Satisfaction

Average rating of 2.99 indicates service quality should remain a focus. Improving consistency during busy periods may increase repeat-order behaviour and loyalty.

8.8 Stakeholder Outcomes

Rahul Sharma: Quick identification of high-demand cities, cuisine/meal preferences, customer segments, and order-value patterns.

Priya Mehta: Workload distribution, city demand, meal-period pressure, and processing-time patterns for resource allocation.

Neha Reddy: Importance of choice, reasonable cost, timely service, and consistent quality for satisfaction and loyalty.

9. Future Scope

The current project provides descriptive analytics using one validated CSV. Value can be extended through:

9.1 Real-Time Data Integration

Connect Tableau to a live database or cloud warehouse with automatic refresh schedules.

9.2 Predictive Demand Forecasting

Forecast demand by city, cuisine, meal type, day type, and time period for staffing and promotions.

9.3 Delivery and Processing-Time Prediction

Estimate expected times using city, meal type, restaurant, weather, and workload.

9.4 Customer Segmentation

Group customers by frequency, spending, preference, discount sensitivity, ratings, and repeat behaviour.

9.5 Profitability Analysis

Include commission, promotion costs, refunds, payment charges, and driver costs.

9.6 Customer Retention Analysis

Measure retention, churn, order frequency, and customer lifetime value.

9.7 Geographic Analysis

Use area, postal code, or coordinates for map-based demand and service analysis.

9.8 Weather and External Factors

Expand weather data; include holidays, festivals, events, and traffic.

9.9 Alerts and Operational Thresholds

Configure Tableau alerts for high processing times, falling ratings, or volume spikes.

9.10 Mobile Optimization

Dedicated mobile layout prioritizing KPIs, alerts, and key operational views.

9.11 Advanced Interaction

Parameters, dynamic titles, set/highlight actions, drill-through, and user-specific views.

9.12 Data-Governance Improvements

Formal data dictionary, refresh checks, version control, and validation rules.

10. Appendix

10.1 Source Code

No separate programming source code was developed. Analysis uses Tableau Desktop, the supplied CSV, Tableau calculations, filters, dashboard objects, and the Food.twb workbook.

10.2 GitHub and Project Demo Link

GitHub: <https://github.com/25eg105h26-code/Food-Ordering-Behaviour-and-Consumer-Trends>

A separate project demo link was not provided.

10.3 Project Files

File or Artifact	Purpose
food_ordering_behavior_dataset.csv	Primary source – 50,000 records, 19 fields
Food.twb	Tableau workbook – calculations, sheets, dashboards, story
Define Problem Statements	Stakeholder problem-definition document
Project Proposal / Planning	Solution, resources, backlog, sprint schedule
Data Quality / Exploration Reports	Quality assessment and preprocessing summary
Business Question & Visualization Report	Business questions and Tableau views
Dashboard Design Activity	Dashboard layout, features, outcomes
Story Design File	Tableau story structure and observations

10.4 Principal Results Summary

Measure or Finding	Result
Dataset records / fields	50,000 / 19
Missing values / Duplicates	0 / 0
Total order value	27.39M
Total delivery fees	2.98M
Average customer rating	2.99
Highest city	Mumbai – 8,443 orders
Highest meal-type order value	Breakfast – 11,83,066
Second-highest meal-type value	Dinner – 11,44,561
Cuisine distribution	~16.6%–16.7% per displayed cuisine
Largest age groups	Millennials and Gen Z

Final Summary

The Food Ordering Behaviour and Consumer Trends project demonstrates how Tableau converts a large, validated dataset into an effective decision-support system. The interactive dashboard and story provide clear views of business scale, city demand, cuisine preferences, customer segments, meal-type performance, ratings, and operational timing.

Findings show strong activity in Mumbai, balanced cuisine demand, notable Millennial and Gen Z participation, and substantial order-value contribution from Breakfast and Dinner. The average rating of 2.99 highlights an opportunity to strengthen service quality. Reliable data preparation, appropriate visualizations, interactive filters, stakeholder scenarios, and structured storytelling support better strategic planning, operational allocation, and customer-experience improvement.